

Series and infinite products – 6

Series in number theory: digits, primes, irrationality

Homework

12 сентября 2026 года

Theoretical minimum. The Euler product is unique factorisation written through geometric progressions; from it one reads the divergence of $\sum 1/p$ with the rate $\ln \ln x$, and the Dirichlet series of d , μ and φ . Restrictions on digits leave geometrically few numbers in each block of length k , which is why Kempner-type series converge. Irrationality of a sum is proved by producing integers with $0 < |q_n x - p_n| \rightarrow 0$: multiply by $n!$ or by a denominator and estimate the tail from both sides; a sum of a rapidly decreasing series is irrational as soon as the tail is smaller than the smallest possible non-zero distance to an integer multiple. A series over an arithmetic function is usually handled by summing along level sets, as in Series and infinite products – 5.

1. (Putnam 2002) Fix an integer $b \geq 2$. Let $f(1) = 1$, $f(2) = 2$, and for each $n \geq 3$ define $f(n) = n f(d)$, where d is the number of base- b digits of n . For which values of b does $\sum_{n=1}^{\infty} \frac{1}{f(n)}$ converge?
2. Let $\sigma_2(n)$ denote the number of ones in the binary representation of a positive integer n . Find the sum of the series $\sum_{n=1}^{\infty} \frac{\sigma_2(n)}{n(n+1)}$.
3. Prove that $e = \sum \frac{1}{n!}$ is irrational, that $\sum \frac{1}{2^{n!}}$ is irrational, and state which of the two arguments also proves transcendence.
4. Prove the identities $\sum \frac{\mu(n)}{n^s} = \frac{1}{\zeta(s)}$ and $\sum \frac{\varphi(n)}{n^s} = \frac{\zeta(s-1)}{\zeta(s)}$ for $s > 2$, and deduce that the density of the squarefree numbers equals $\frac{6}{\pi^2}$.
5. (CIIM 2024) Given a real number x , define $S(x) = \sum_{n=1}^{\infty} \{n! \cdot x\}$, where $\{s\}$ is the fractional part. Determine whether there exists an irrational x for which $S(x)$ converges.

6. Prove Kempner's theorem in base b for an arbitrary forbidden digit, and determine for which b and which digit the resulting sum is the largest.
7. Prove that $\sum_p \frac{1}{p \ln p}$ converges, using only the bound $\sum_{p \leq x} \frac{1}{p} = \ln \ln x + O(1)$ and summation by parts.
8. Prove Euler's elementary lower bound $\sum_{p \leq x} \frac{1}{p} \geq \ln \ln x - C$ without using the prime number theorem, and explain why the same argument gives the infinitude of primes in a stronger form than Euclid's.
9. Prove that if positive integers satisfy $\frac{a_{n+1}}{a_n^2} \rightarrow \infty$, then $\sum \frac{1}{a_n}$ is irrational. Show that under the weaker condition $a_{n+1} \geq a_n^2 - a_n + 1$ the sum may be rational: for Sylvester's sequence $2, 3, 7, 43, 1807, \dots$ it equals 1.
10. Prove that $\sum_{n \geq 1} \frac{1}{\text{lcm}(n, n+1)}$ converges and compute it; then prove that $\sum_{n \geq 1} \frac{1}{\text{lcm}(1, 2, \dots, n)}$ converges, and estimate how fast.